CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester Master of Technology (CE)
University Examination, January 2010

CE701 Operating System Design and Concepts

Date: 03/01/2011 **Time: 10:30 a.m. to 1:30 p.m.** **Maximum Marks: 70**

Instructions:
1. Figure to the right indicates full marks
2. Make suitable assumption whenever necessary and mention it clearly
3. Parts of same question should be answered together and in the same sequence.
4. Answers to the sections must be written separately.

SECTION-I

Q-1

(a) Define system response time as the average time it takes to complete system call. Define system throughput as the number of process the system can execute in a given time period. Describe how the buffer cache can help response time. Does it necessarily help system throughput? **4**

(b) How a kernel interface and a user-level thread package that together combine the functionality of kernel threads with the performance and flexibility of user-level threads. **3**

(c) How does UNIX kernel maintain the consistency of the system data structures? **2**

(d) Differentiate between close() and shutdown() in terms of socket communication. **2**

Q-2

(a) When a resource becomes available, the wakeup() routine wakes up all processes blocked on it. What are the drawbacks of this approach? What are the alternatives? **4**

(b) Explain about modular kernel. **2**

(c) How UNIX kernel maintains buffers in free list of buffer cache? "Buffer allocation algorithm is safe ; process must not sleep forever and they must eventually get a buffer" Justify the statement **6**

OR

(b) Can a shared ready queue limit the scalability of a multiprocessor system? **4**

(c) Describe a scenario where the buffer data is already valid in algorithm bread. **4**

Q-3 **Answer the following questions. (Any Three)** **12**

(a) The proc structure and the u area contain process attributes and resources. In a multithreaded system, which of their fields may be shared by all LWPs of the process, and which must be per-LWP?

(b) Discuss virtual I/O in Unix.

(c) What must a system support to have a minimal DSM system?

(d) What is stripping? What are its advantages and disadvantages?

SECTION-II

Q-4

(a)	In UNIX, accessing larger file is expensive than smaller file. Justify	**2**
(b)	Why dummy process is required in multiprocessor architecture?	**2**
(c)	Write an algorithm for pipe system call.	**2**
(d)	How is list of free blocks maintained in UNIX File System?	**2**
(e)	Define in-core inode. Which information is stored in in-core i-node?	**3**

Q-5

(a)	What are the system calls associated with Shared Memory? Explain functionality of each system call.	**4**
(b)	Draw the architecture of Satellite processor distributed system. How fork() system call is executed in a central and a satellite processor.	**8**
	OR	
(b)	Write & Explain an algorithm for namei which converts pathname to inode.	**8**

Q-6	**Answer the following questions. (Any three)**	**12**
(a)	Briefly explain the directory structure in UNIX. Why the inode does starts from 1, not from 0?	
(b)	What is race condition associated with unlink system call? What is solution of it?	
(c)	What are the entries of kernel mount table? Write an algorithm for mount system call.	
(d)	What are the primary features of reliable signal?	